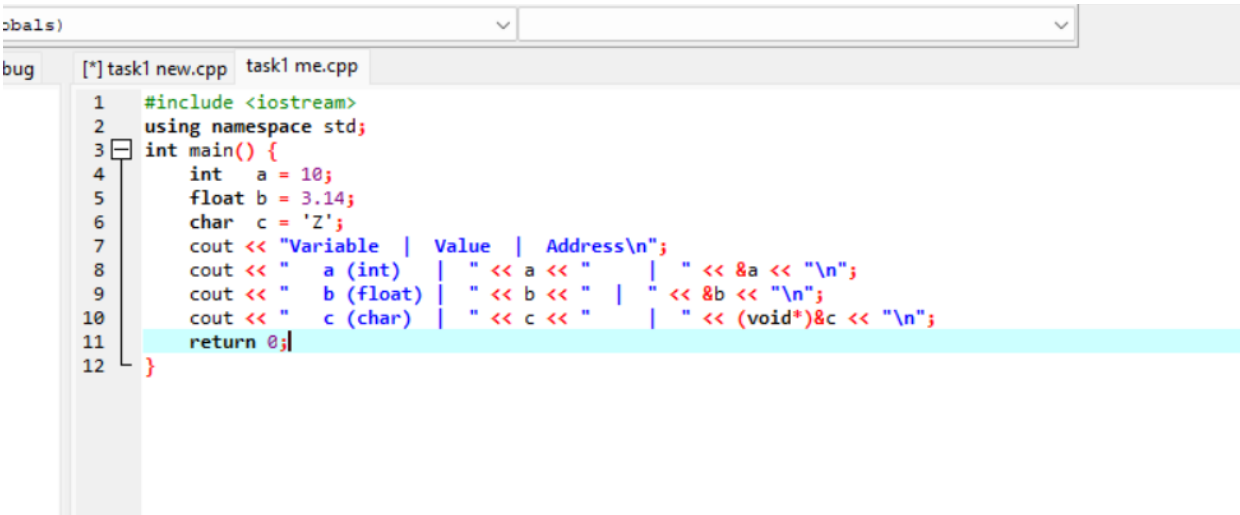
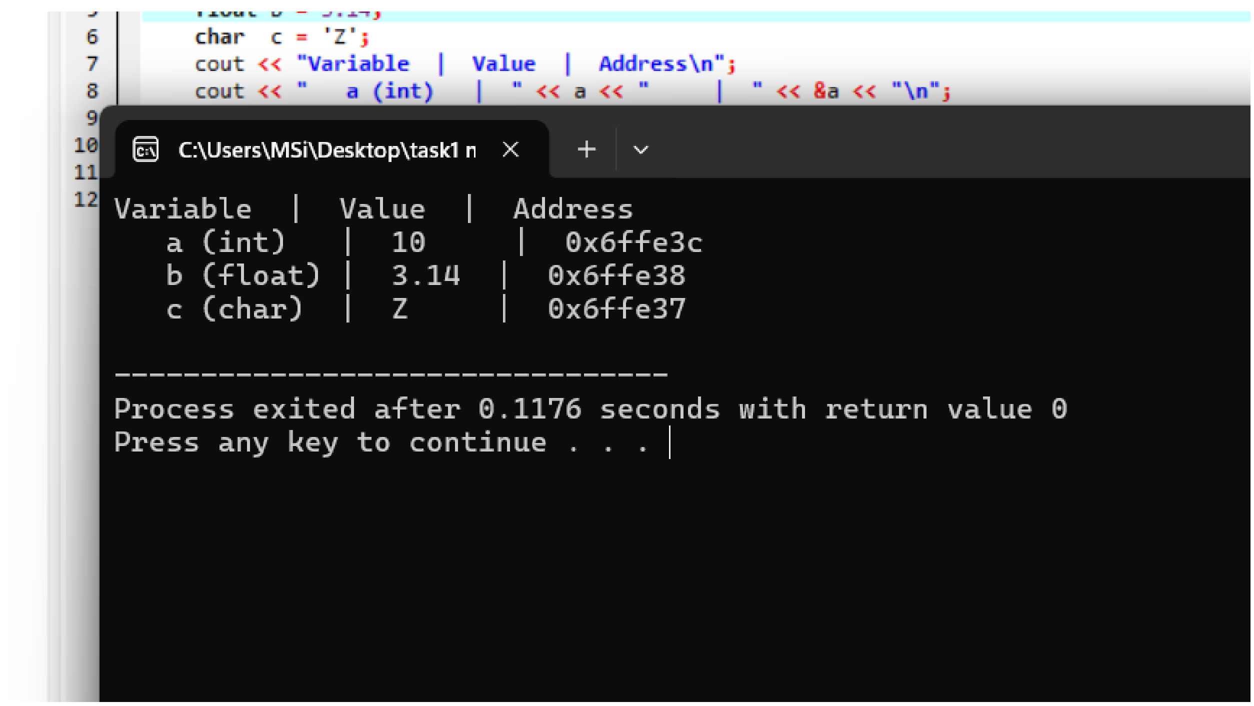
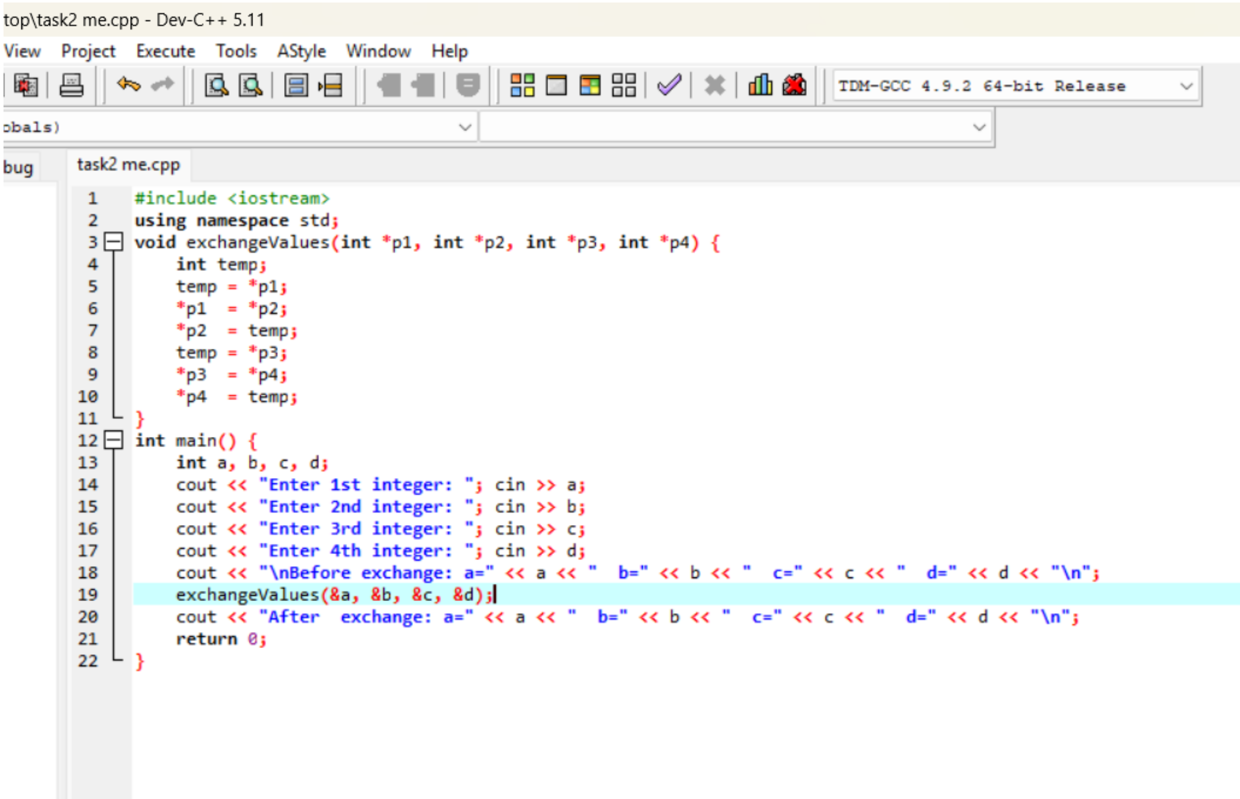
# REPORT DOCUMENT

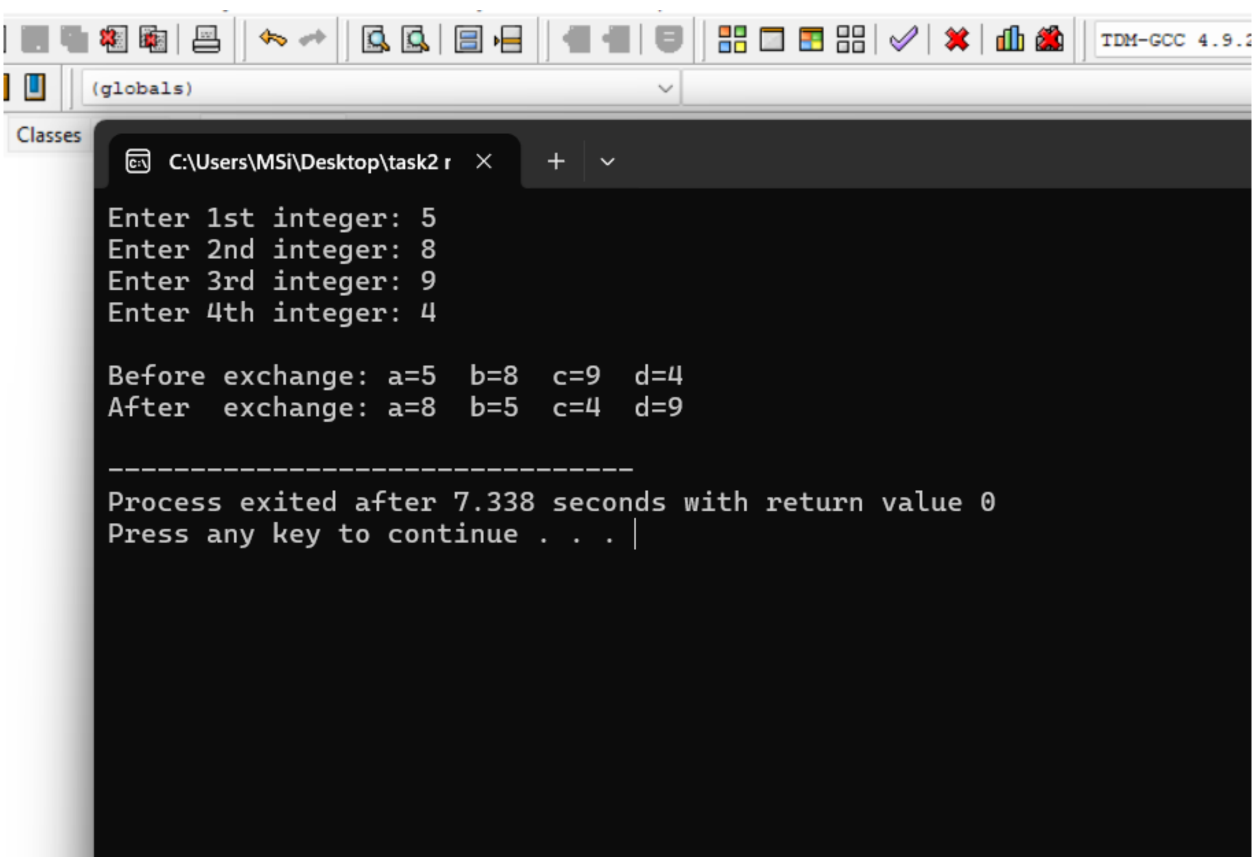
## Lab Task 1:




## Lab Task 2


top\task2 me.cpp - Dev-C++ 5.11
View Project Execute Tools AStyle Window Help
TDM-GCC 4.9.2 64-bit Release
task2 me.cpp
#include <iostream>
using namespace std;
void exchangeValues(int *p1, int *p2, int *p3, int *p4) {
int temp;
temp = *p1;
*p1 = *p2;
*p2 = temp;
temp = *p3;
*p3 = *p4;
*p4 = temp;
}
int main() {
int a, b, c, d;
cout << "Enter 1st integer: "; cin >> a;
cout << "Enter 2nd integer: "; cin >> b;
cout << "Enter 3rd integer: "; cin >> c;
cout << "Enter 4th integer: "; cin >> d;
cout << "\nBefore exchange: a=" << a << "  b=" << b << "  c=" << c << "  d=" << d << "\n";
exchangeValues(&a, &b, &c, &d);
cout << "After  exchange: a=" << a << "  b=" << b << "  c=" << c << "  d=" << d << "\n";
return 0;
}


(globals)
Classes
C:\Users\MSi\Desktop\task2 r
Enter 1st integer: 5
Enter 2nd integer: 8
Enter 3rd integer: 9
Enter 4th integer: 4
Before exchange: a=5  b=8  c=9  d=4
After  exchange: a=8  b=5  c=4  d=9
--------------------------------
Process exited after 7.338 seconds with return value 0
Press any key to continue . . .

# Lab Task 3

```cpp
#include <iostream>
using namespace std;
void inputArray(int *arr, int size) {
    cout << "Enter " << size << " values:\n";
    for (int i = 0; i < size; i++) {
        cout << "  Element [" << i << "]: ";
        cin >> *(arr + i);
    }
}
int main() {
    const int SIZE = 5;
    int arr[SIZE];
    inputArray(arr, SIZE);
    cout << "\nArray values are:\n  ";
    for (int i = 0; i < SIZE; i++) {
        cout << arr[i];
        if (i < SIZE - 1) cout << "  ";
    }
    cout << "";
    return 0;
}
```

```
Enter 5 values:
  Element [0]: 2
  Element [1]: 5
  Element [2]: 7
  Element [3]: 9
  Element [4]: 9

Array values are:
  2  5  7  9  9

--------------------------------
Process exited after 14.16 seconds with return value 0
Press any key to continue . . .
```

<u>Lab Task 4</u>

Untitled1 | task4 me org.cpp

```cpp
#include <iostream>
using namespace std;
void Largest(int *a, int *b, int *c) {
    int largest = *a;
    if (*b > largest) largest = *b;
    if (*c > largest) largest = *c;
    cout << "Largest  = " << largest << "\n";
}
void Smallest(int *a, int *b, int *c) {
    int smallest = *a;
    if (*b < smallest) smallest = *b;
    if (*c < smallest) smallest = *c;
    cout << "Smallest = " << smallest << "\n";
}
void Average(int *a, int *b, int *c) {
    double avg = (*a + *b + *c) / 3.0;
    cout << "Average  = " << avg << "\n";
}
int main() {
    int x, y, z, choice;
    cout << "Enter three integers:\n";
    cout << "  Number 1: "; cin >> x;
    cout << "  Number 2: "; cin >> y;
    cout << "  Number 3: "; cin >> z;
    do {
        cout << "\n-------- MENU --------\n";
        cout << "1. Find Largest\n";
        cout << "2. Find Smallest\n";
        cout << "3. Find Average\n";
        cout << "4. Exit\n";
        cout << "Enter choice: ";
```

```
Enter three integers:
  Number 1: 3
  Number 2: 5
  Number 3: 7

-------- MENU --------
1. Find Largest
2. Find Smallest
3. Find Average
4. Exit
Enter choice: 3
Average  = 5
```

- HOME TASKS

Home Task 1

```cpp
#include<iostream>
using namespace std;
int maxNum(int* ptr, int size)
{
    int max = *ptr;
    for(int i = 1; i < size; i++)
    {
        if(*(ptr + i) > max)
            max = *(ptr + i);
    }
    return max;
}
int main()
{
    int arr[] = {34, 7, 89, 12, 56, 43};
    int size = sizeof(arr) / sizeof(arr[0]);
    cout << "Maximum: " << maxNum(arr, size) << endl;
    return 0;
}
```

```
C:\Users\MSi\Desktop\home
Maximum: 89

--------------------------------
Process exited after 0.1221 seconds with return value 0
Press any key to continue . . .
```

Home task 2

home task 1 me.cpp | home task 2 me.cpp

```cpp
#include<iostream>
using namespace std;
void arrReverse(int* ptr, int size)
{
    int* start = ptr;
    int* end = ptr + size - 1;
    while(start < end)
    {
        int temp = *start;
        *start = *end;
        *end = temp;
        start++;
        end--;
    }
}
int main()
{
    int arr[] = {1, 2, 3, 4, 5};
    int size = 5;
    arrReverse(arr, size);
    for(int i = 0; i < size; i++)
        cout << arr[i] << " ";
    return 0;
}
```

```
5 4 3 2 1
--------------------------------
Process exited after 0.11 seconds with return value 0
Press any key to continue . . .
```

## Home task 3

```cpp
#include <iostream>
using namespace std;
int countVowels(char* str) {
    char* ptr = str;
    int count = 0;
    int index = 0;
    cout << "\nVowels found:\n";
    while (*ptr != '\0') {
        char ch = *ptr;
        if (ch=='a' || ch=='e' || ch=='i' || ch=='o' || ch=='u' ||
            ch=='A' || ch=='E' || ch=='I' || ch=='O' || ch=='U') {
            cout << "  Vowel: '" << ch << "'  at index " << index << "\n";
            count++;
        }
        ptr++;
        index++;
    }
    return count;
}
int main() {
    char str[]="Programming";
    int total = countVowels(str);
    cout << "\nTotal vowels = " << total << "\n";
    return 0;
}
```

```
C:\Users\MSi\Desktop\extens

Vowels found:
  Vowel: 'o'  at index 2
  Vowel: 'a'  at index 5
  Vowel: 'i'  at index 8

Total vowels = 3

--------------------------------
Process exited after 0.1372 seconds with return value 0
Press any key to continue . . .
```

## Home task 4

home task 4 me.cpp

```cpp
#include<iostream>
using namespace std;
void StringCopy(char* dest, char* src)
{
    while(*src != '\0')
    {
        *dest = *src;
        dest++;
        src++;
    }
    *dest = '\0';
}
int main()
{
    char source[] = "Hello, Pointers!";
    char destination[50];
    StringCopy(destination, source);
    cout << "Copied: " << destination << endl;
    return 0;
}
```

```
Copied: Hello, Pointers!

--------------------------------
Process exited after 0.1226 seconds with return value 0
Press any key to continue . . .
```

## Home Task 5

home task 5 me.cpp

```cpp
#include<iostream>
using namespace std;
void bubbleSort(int* arr, int n)
{
    for(int i = 0; i < n - 1; i++)
    {
        for(int j = 0; j < n - i - 1; j++)
        {
            if(*(arr + j) > *(arr + j + 1))
            {
                int temp = *(arr + j);
                *(arr + j) = *(arr + j + 1);
                *(arr + j + 1) = temp;
            }
        }
    }
}
int main()
{
    int arr[] = {63, 24, 11, 21, 10};
    int n = 5;
    cout << "Before: ";
    for(int i = 0; i < n; i++)
        cout << *(arr + i) << " ";
    bubbleSort(arr, n);
    cout << "\nAfter: ";
    for(int i = 0; i < n; i++)
        cout << *(arr + i) << " ";
    return 0;
}
```

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```
Before: 63 24 11 21 10
After: 10 11 21 24 63
--------------------------------
Process exited after 0.1331 seconds with return value 0
Press any key to continue . . .
```

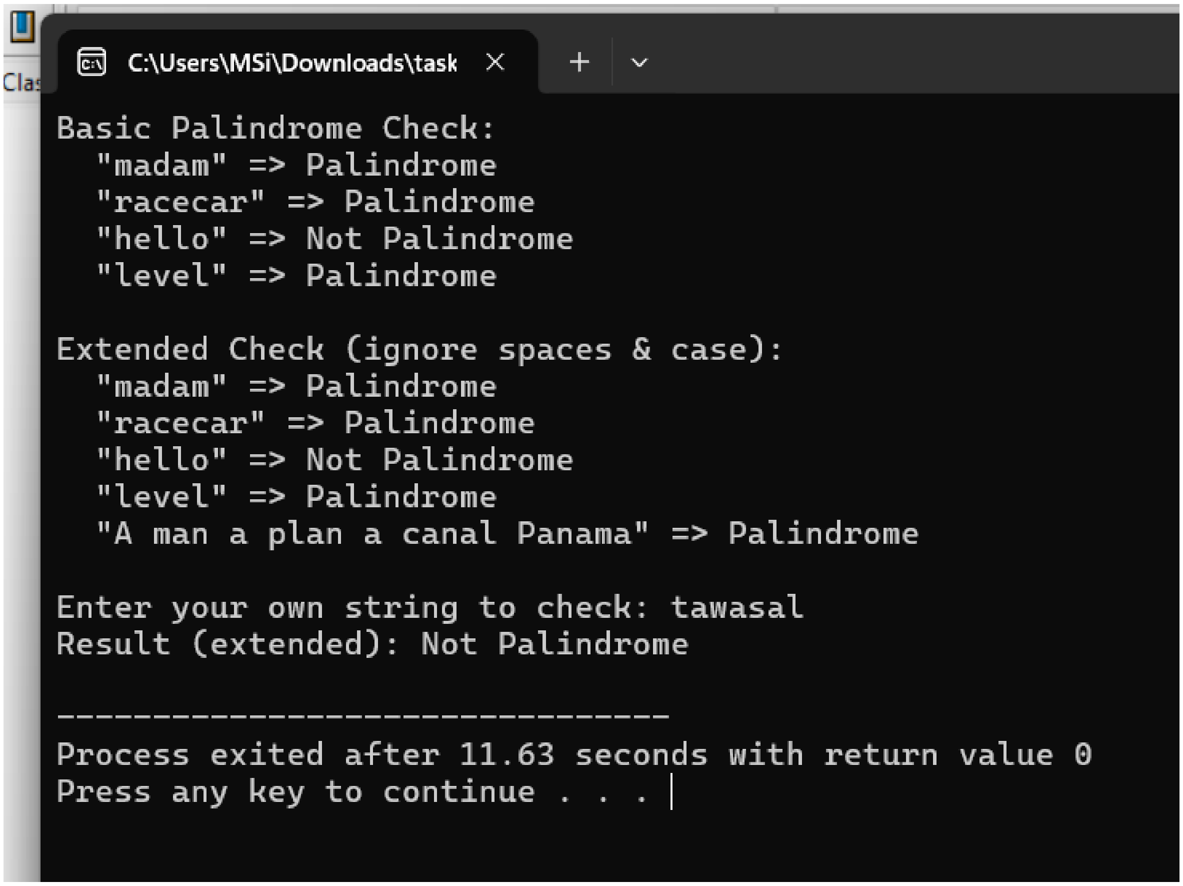
## Home Task 6

[*] task6_palindrome.cpp

```cpp
#include <iostream>
using namespace std;
int myStrLen(char* s) {
    char* p = s;
    while (*p != '\0') p++;
    return (int)(p - s);
}
bool isPalindrome(char* s) {
    int len   = myStrLen(s);
    char* left  = s;
    char* right = s + len - 1;
    while (left < right) {
        if (*left != *right) return false;
        left++;
        right--;
    }
    return true;
}
char toLow(char c) {
    if (c >= 'A' && c <= 'Z') return c + 32;
    return c;
}
bool isLetter(char c) {
    return (c >= 'a' && c <= 'z') || (c >= 'A' && c <= 'Z');
}
bool isPalindromeExtended(char* s) {
    int len   = myStrLen(s);
    char* left  = s;
    char* right = s + len - 1;

    while (left < right) {

```




<u>Home Task 7</u>

```cpp
#include <iostream>
using namespace std;
int main() {
    int rows, cols;
    cout << "Enter number of rows    : "; cin >> rows;
    cout << "Enter number of columns: "; cin >> cols;
    int** matrix = new int*[rows];
    for (int i = 0; i < rows; i++) {
        matrix[i] = new int[cols];
    }
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < cols; j++) {
            matrix[i][j] = i * cols + j + 1;
        }
    }
    cout << "\nMatrix (" << rows << " x " << cols << "):\n";
    for (int i = 0; i < rows; i++) {
        cout << "  ";
        for (int j = 0; j < cols; j++) {
            cout << matrix[i][j];
            if (j < cols - 1) cout << "\t";
        }
        cout << "\n";
    }
    for (int i = 0; i < rows; i++) {
        delete[] matrix[i];
    }
    delete[] matrix;
    cout << "\nMemory deallocated successfully.\n";
    return 0;
}
```

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```
Enter number of rows    : 3
Enter number of columns: 2

Matrix (3 x 2):
  1       2
  3       4
  5       6

Memory deallocated successfully.
```

## Home Task 8

```cpp
#include<iostream>
using namespace std;
void Prime(int *num)
{
    int count = 0;
    for(int i = 1; i <= *num; i++)
    {
        if(*num % i == 0)
        {
            count++;
        }
    }
    if(count == 2)
        cout << *num << " is Prime Number";
    else
        cout << *num << " is Not Prime Number";
}
int main()
{
    int a;
    cout << "Enter any integer: ";
    cin >> a;
    Prime(&a);
    return 0;
}
```

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```
Enter any integer: 3
3 is Prime Number
--------------------------------
Process exited after 2.773 seconds with return value 0
Press any key to continue . . .
```

## Home Task 9

```cpp
using namespace std;
void inputArray(int *arr, int size)
{
    cout << "Enter "<< size <<" values:\n";
    for(int i = 0; i < size; i++)
    {
        cin >> *(arr + i);
    }
}
void doubArray(int *arr, int size)
{
    for(int i = 0; i < size; i++)
    {
        *(arr + i) = *(arr + i) * 2;
    }
}
int main()
{
    int arr[3];
    inputArray(arr, 3);
    cout << "Original Array: ";
    for(int i = 0; i < 3; i++)
    {
        cout << *(arr + i) << " ";
    }
    doubArray(arr, 3);
    cout << "\nDoubled Array(new): ";
    for(int i = 0; i < 3; i++)
    {
        cout << *(arr + i) << " ";
    }
```

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```
Enter 3 values:
1 2 3
Original Array: 1 2 3
Doubled Array(new): 2 4 6
--------------------------------
Process exited after 2.545 seconds with return value 0
Press any key to continue . . .
```